COMPUTER NETWORKS

Date:

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ASSIGNMENT 02

TASK 01:

1. Why is reliable Data Transfer important in this Network?

- Reliable Data Transfer is critical because field rescue teams and the Central Emergency Operation Center (EOC) exchange vital casualty list, supply items and rescue assignments. A lost or corrupted packet could easily misdirect a rescue team to the wrong physical location. Therefore, that every packet arrives correctly and in order is essential for life saving disaster response operation.

2. Why can Stop-And-Wait be inefficient on this link?

- The satellite link features a high RTT (500ms) paired with a packet size 1000bytes over 2MBps. The packet transmission time is extremely short meaning the sender remains idle for the vast majority of RTT waiting for an acknowledgment. This result in very low channel utilization, severely restricting the speed of Emergency communication.

3. Why is pipelining useful here?

- Pipeline allows sender to transmit multiple packets consecutively without waiting for an acknowledgment. Give the high RTT and limited Bandwidth(2MBps) of satellite Link, pipelining keeps the transmission channel busy. This drastically increases link utilization and throughput, enabling faster transmission of urgent rescue data.

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4. Is GBN or SR more suitable for this scenario?

- Selective Repeat (SR) is more suitable than Go-Back-N (GBN) for this constrained link:
  - Bandwidth conservation: The link has limited bandwith and suffers from Packet loss and corruption SR only retransmitts the specific lost or corrupted packets. whereas GBN would waste precious bandwidth by retransmission of an entire window of successfully received packets.
  - Overhead Reduction: With a [redacted] 500ms RTT retransmission of large blocks of packets under GBN introduces massive delay penalties, making SR's individual packet recovery mechanism more efficient for maintaining throughput.

TASK 02:

Packets: 10 (0 - 9)

Window Size : 04

Loss Event: Packet 02 is lost ; packet 03 and 04 reach the receiver end. All ACK of packets arrives correctly in order.

Timeout: Timeout for packet 02 occurs before packet 02 is successfully received again.

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a) GO-Back-N (GBN) Packet Trace

Event	Sender	Packet	Receiver	ACK	Window
1	Send Pkt0	0	Recv pkt0, delivers	Ack0	[0-3]
2	Send Pkt 1	1	Recv pkt0, delivers	Ack1	[0-3]
3	Send Pkt 2	2	-(Not Received)	-	[0-3]
4	Send Pkt3	3	Recv Pkt 3 outof order, discard	Ack1 (duplicate)	[0-3]
5	Send Pkt 4	4	Recv Pkt 4 Outof Order, discard	Ack 1 (duplicate)	[0-3]
6	Receives Ack 0	-	-	-	[1-4]
7	Receives Ack 1	-	-	-	[2-5]
8	Timeout for Pkt 2	-	-	-	[2-5]
9	Retransmit Pkt 2	2	Recv Pkt 2 In-order	Ack 4	[2-5]
10	Send Pkt 5	5	Recv Pkt 5 deleivers	Ack 5	[5-8]
11	Send Pkt 6	6	Recv Pkt 6 delivers	Ack 6	[6-9]
12	Send Pkt 7	7	Recv Pkt 7 delivers	Ack 7	[7,9],[0
13	Send Pkt 8	8	Recv Pkt 8 delivers	Ack 8	[8-9],[0,[
14	Send Pkt 9	9	Recv Pkt 9 delivers	Ack 9	[9],[0,2]

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b) Selective Repeat (SR) Packet Trace

Event	Sender	Packet	Receiver	ACK	Window
1	Send Pkt 0	0	Recv Pkt 0 delivers	ACK 0	[0-3]
2	Send Pkt 1	1	Recv Pkt 1	ACK 1	[0-3]
3	Send Pkt 2	2	– (Not Received)	–	[0-3]
4	Send Pkt 3	3	Recv Pkt 3, out of order buffers, send Ack 3	ACK 3	[0-3]
5	Send Pkt 4	4	Recv Pkt 4, out of order buffers, Send Ack 4	ACK 4	[0-3]
6	Recv ACK 0	–	–	–	[1-4] base moves to 1
7	Recv ACK 1	–	–	–	[2-5] base moves to 2
8	Recv ACK 3 (Buffers)	–	–	–	base stays at 2
9	Recv ACK 4 (Buffers)	–	–	–	base stays at 2
10	Timeout Pkt 2	–	–	–	[2-5]
11	Retransmit Pkt 2	2	Recv Pkt 2, fills gap, delivers 2,3,4	ACK 2	[5-8] base moves to 5
12	Send pkt 5	5	Recv Pkt 5 delivers	ACK 5	[6-9]
13	Send Pkt 6	6	Recv Pkt 6, delivers	ACK 6	[7,9],[0]
14	Send Pkt 7	7	Recv Pkt 7, delivers	ACK 7	[8-9],[0,1]
15	Send Pkt 8	8	Recv Pkt 8, delivers	ACK 8	[9],[0-2]
16	Send Pkt 9	9	Recv Pkt 9, delivers	ACK 9	[0-3]

Completed.

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Summary:

- Number of Retransmitted:
  - GBN: 3 packets (2, 3, 4 were retransmitted because GBN discards out-of-order packets)
  - SR: 1 Packet (Only Pkt retransmitted as SR keep the outoforder packet in buffer).
- They differ because of how the receiver handles out of order packets. GBN uses a strict in-order policy were any packet arriving after a gap is discarded, require the sender to retransmit entire window. In contrast, SR accepts and buffers out-of-order packets, allowing individual acknowledgments, and ensuring that only the specific lost packet is transmitted.

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Task 05:

1- **Pipelining:** Stop-and-wait forces the sender to halt after transmitting a single packet and remain idle until the acknowledgment returns, which severely wastes bandwidth on link with a high RTT. Pipelining allows the sender to transmit multiple packets back-to-back without waiting for immediate ACKs, keeping the pipeline full and increases link utilization

Transmission Time $= \frac{1}{R} = \frac{8000}{2000000} = 0.004 \Rightarrow 4ms.$

Stop & wait Utilization

$$U = \frac{L/R}{RTT + L/R} \Rightarrow \frac{0.004}{0.5 + 0.004} \Rightarrow 0.0079 \text{ or } 0.79\%$$

with $N = 4$

$$U = \frac{N \cdot L/R}{RTT + L/R} = 4 \times 0.0079 \Rightarrow 0.0317 \text{ or } 3.17\%$$

2. a) with 3 bits, $2^3 \Rightarrow 8$ available sequence numbers.

b) Maximum window size (N) must satisfy the condition $N \leq 2^{k-1}$, where k is Number of Seq. bit.

with $k = 3$

$$N \leq 2^{3-1} \Rightarrow 2^2 = 4$$

Max Safe window = 4

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